

Isla Stiven

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EDUCATION

Bachelor of Engineering in Mechanical Engineering (B.Eng)

University of Victoria | GPA: 3.7

Expected Graduation: December 2026

Victoria, BC

SKILLS

Design Skills	CAD Design for Manufacturing FEA CFD Rapid Prototyping PCB Design GD&T Instrumentation Engineering Drawing Shop Tools
Technologies	SolidWorks Siemens NX ANSYS Granta EduPack KiCAD LabVIEW MATLAB Python C++

EXPERIENCE

Telix ARTMS

January 2025 – August 2025

Mechatronics / Mechanical Engineering Co-op

Burnaby, BC

- **Designed and prototyped** a device to measure internal component temperatures during **cyclotron irradiation**, engineered to meet strict requirements such as cyclotron interface compatibility, a fully sealed enclosure, and zero-interference with the irradiation process to improve understanding of **thermal failure modes**.
- Designed, assembled, and tested a **printed circuit board (PCB)** for ARTMS's **nickel-electroplating unit** used in the manufacturing of solid targets for **radioisotope production**.
- Employed **geometric dimensioning and tolerancing (GD&T)** on engineering drawings to define part tolerances, ensuring clear communication with **machinists**.
- Designed, conducted, and documented **experimental tests** to improve isotope production processes, including **water-cooling systems** with **Swagelok** fittings, target polishing methods, and **radiation-hard material selection**.
- **Assembled and integrated** multiple major subsystems of the ARTMS **GEN2 QIS**, deployed on cyclotrons to increase yield and improve efficiency of radioisotope production at **international sites**.

University of Victoria SEGAL Lab

September 2023 – December 2023

Software Engineering Co-op

Victoria, BC

- **Designed and built** a cross-platform educational mobile app prototype in **Flutter** within a **start-up environment** to assist schoolteachers in remote communities.
- Planned project sprints, **led meetings**, and presented at multiple **university conferences** which led to an interview with CBC and an invite to the AISES National Conference.

PROJECTS

Wingfoil Handle Force Measurement System

January 2026 – Present

Tagmata Materials

Victoria, BC

- Designed a **force measurement system** for wing foil athletes using **dual inline load cells** mounted to a **carbon fiber** auxiliary boom, enabling independent front and rear hand force capture during live water sessions.
- Developed a **waterproof, salt-resistant electronics package** incorporating an **ESP32-S3 microcontroller**, dual ADC amplifiers, and a **LiPo power system**, housed within a sealed enclosure for reliable operation in saltwater environments.

UVic FSAE Suspension A-Arm Node Design

February 2026 – April 2026

University of Victoria Formula Racing

Victoria, BC

- Designed four aluminum suspension A-arm nodes in **SolidWorks**, using **FEA**, **topology optimization**, and analytical hand calculations to achieve **64–70% mass reduction** per component while meeting safety factor requirements.
- Applied **design-for-manufacturing** principles to translate optimized geometries into **CNC-machinable** aluminum nodes bonded to **carbon fiber tubes**, sizing the adhesive joints to carry suspension loads simulated in Dynatune.

HOBBIES

Running, Hiking, Piano, Cycling, Volleyball